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Vermeulen

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(54) **REAR FIREARM SIGHT SLIDE ELEVATOR**
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F41G 1/28 (2006.01)
(52) **U.S. Cl.**
CPC **F41G 1/28** (2013.01)
(58) **Field of Classification Search**
CPC F41G 1/28; F41G 1/16
USPC 42/136
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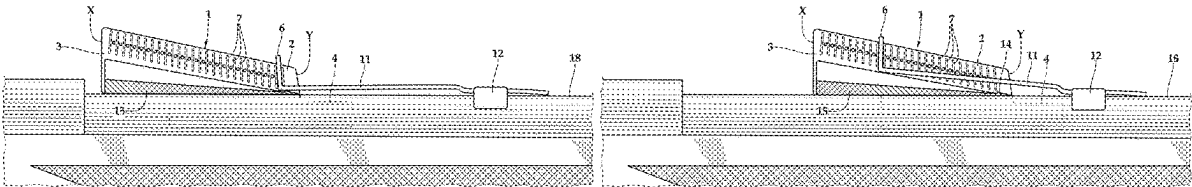
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(57) **ABSTRACT**

A rear firearm sight slide elevator with a base member that is configured to surround part of a firearm barrel and has a first wing on one side and a second wing on another side of the front part of the base member, first and second fins that are parallel to each other and situated at least in part on top of the base member, and a smooth ramp that is situated on a central portion of the base member between the fins. The ramp increases in height from front to back and has a central channel that is configured to provide a sight path from the user to the front sight on the firearm. The inner wall of the first and/or second fin includes a plurality of vertically oriented ridges. The outer wall of at least one of the first and second fins displays a plurality of sight markings.

13 Claims, 7 Drawing Sheets



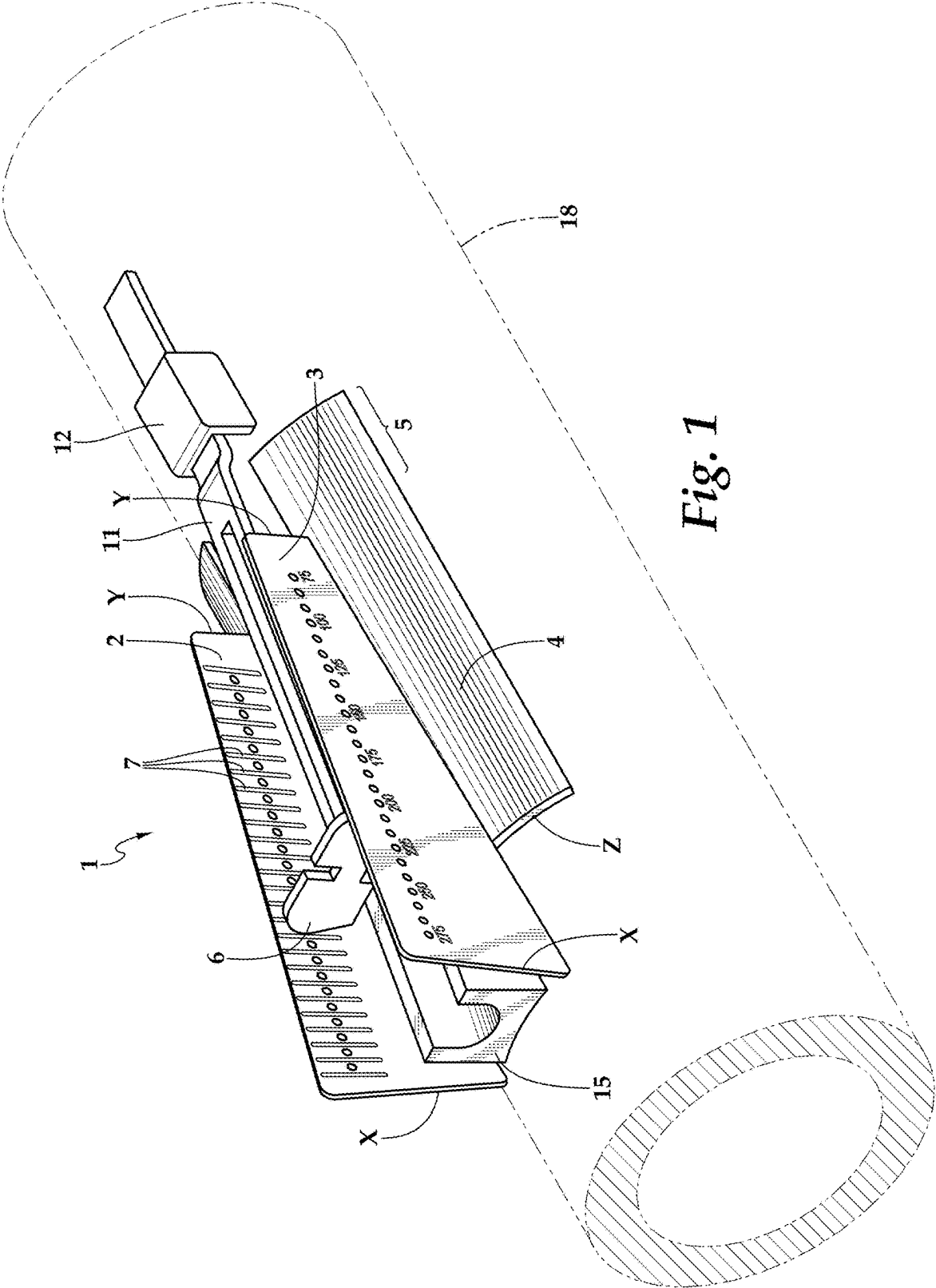
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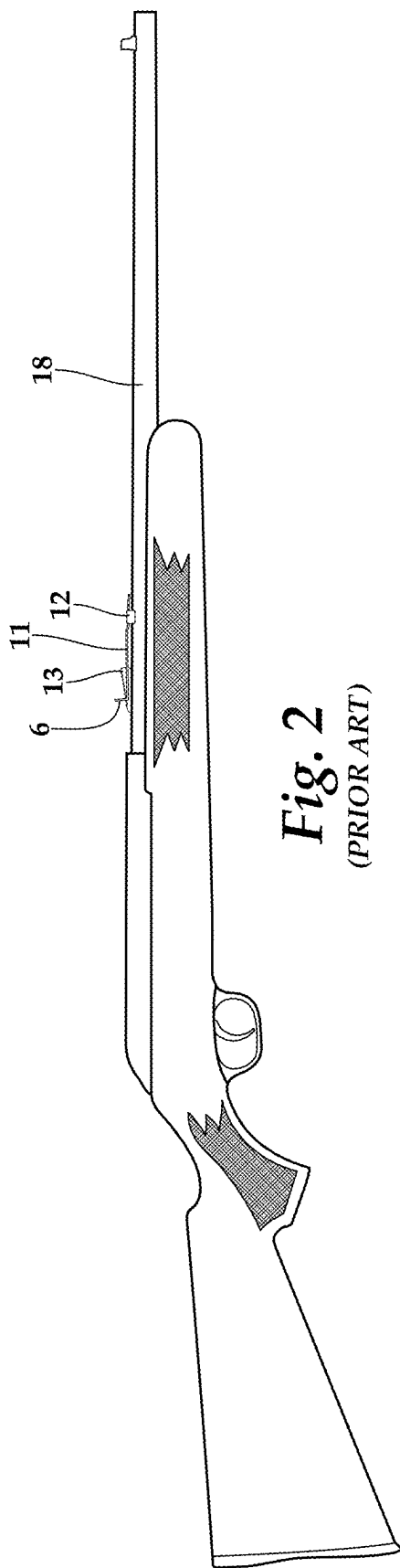


Fig. 2
(PRIOR ART)

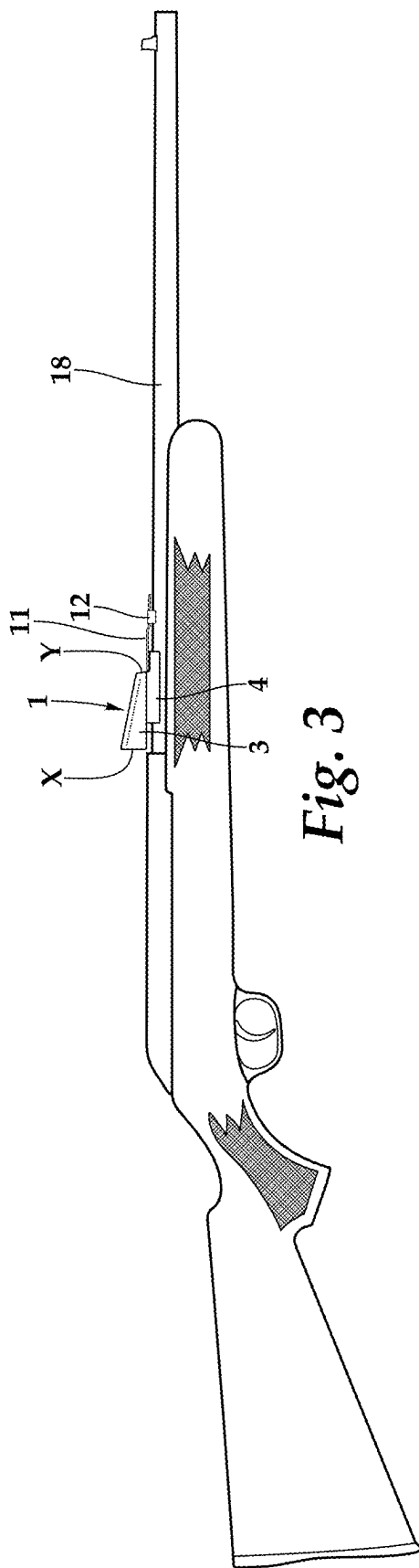


Fig. 3

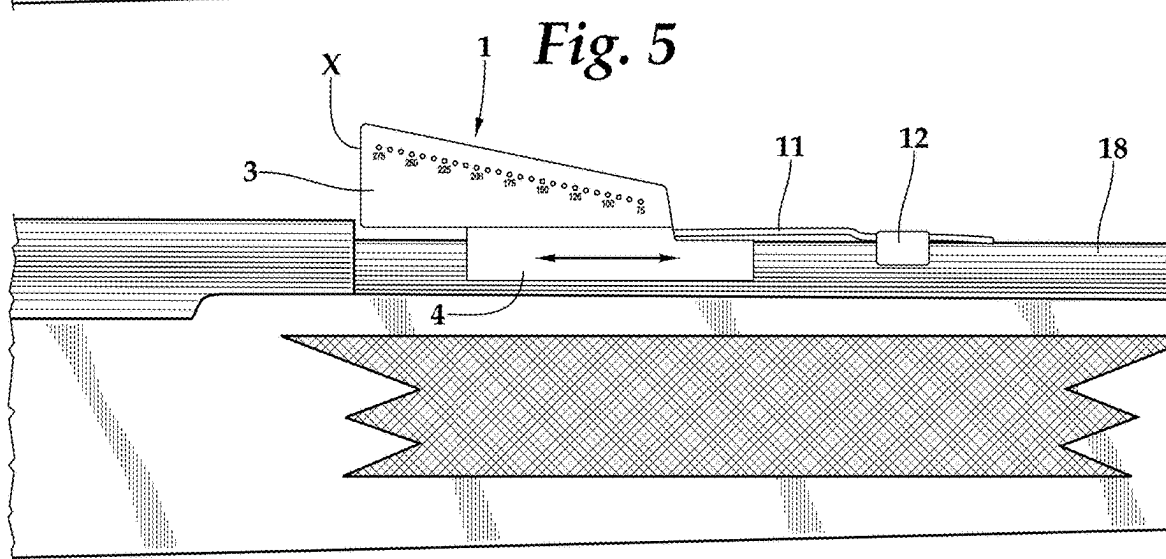
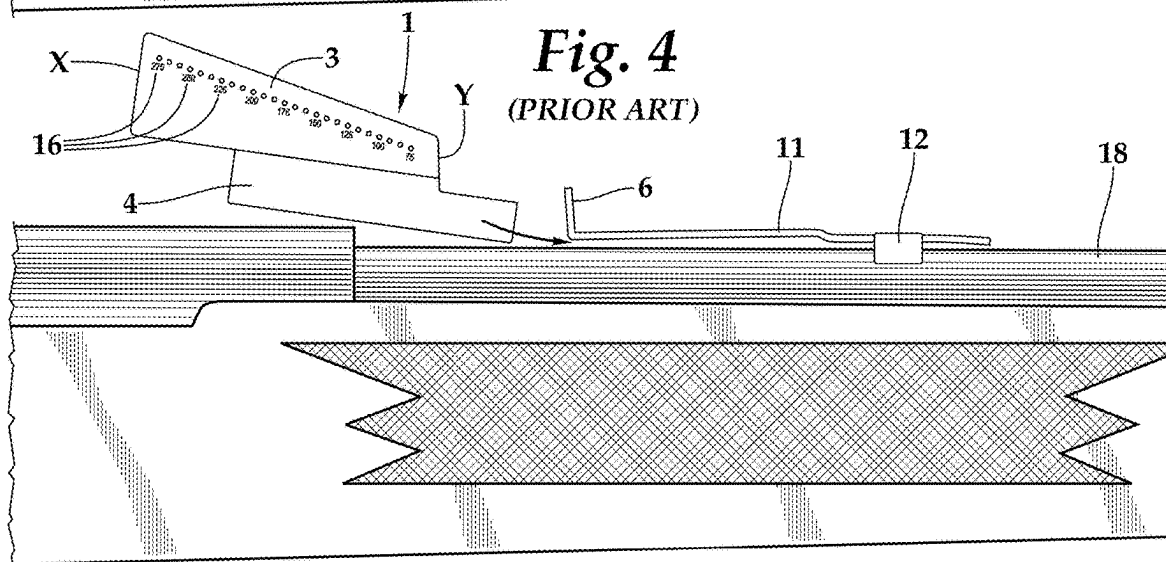
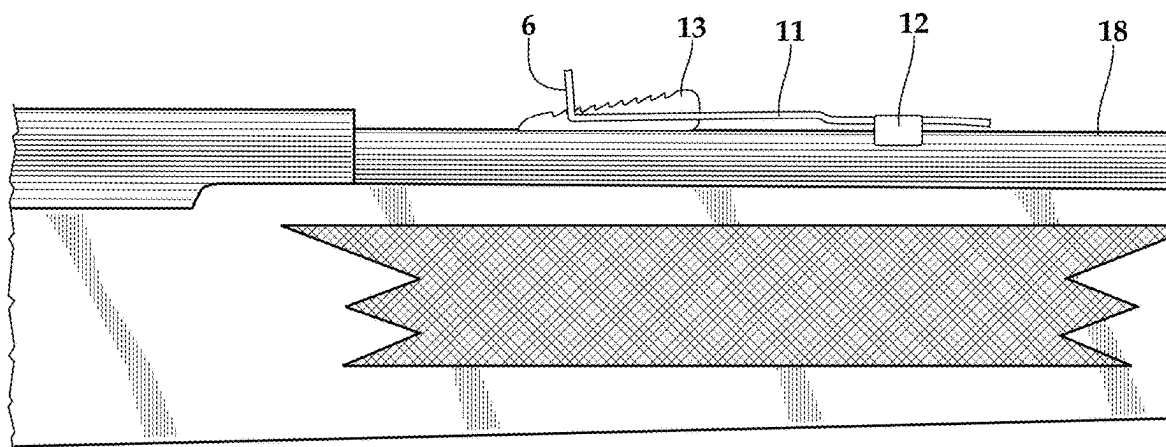
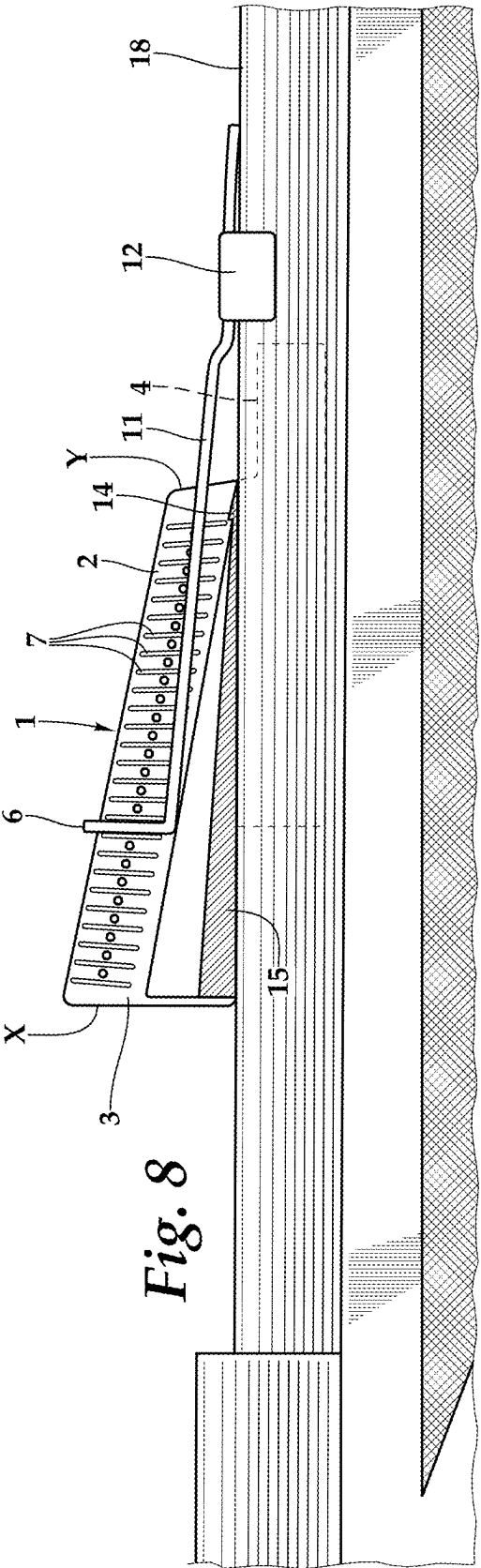
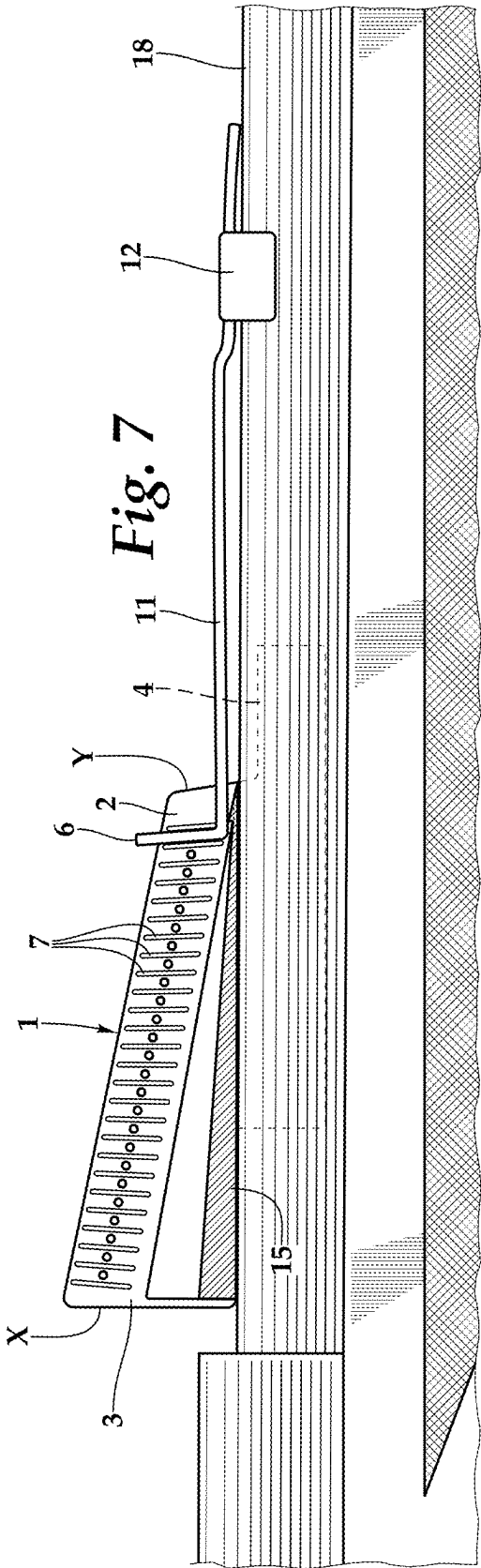


Fig. 6



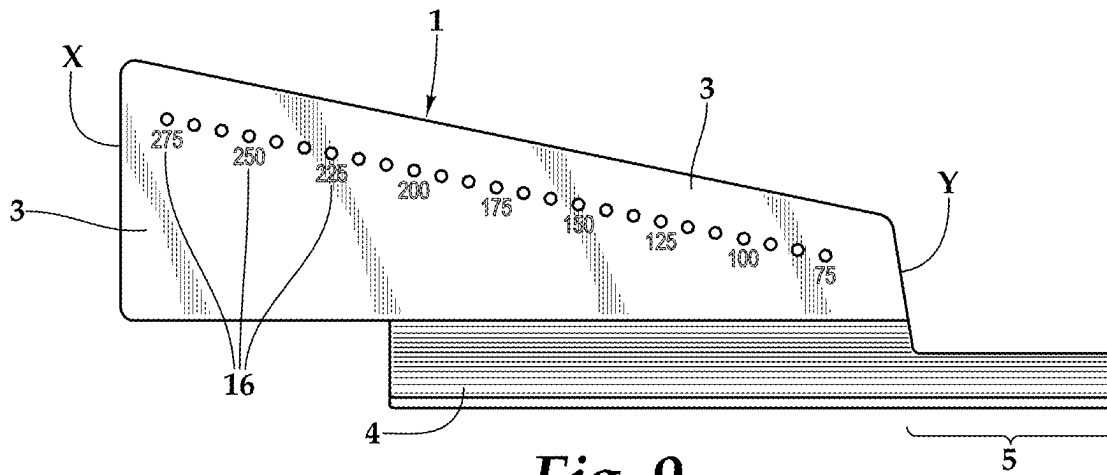


Fig. 9

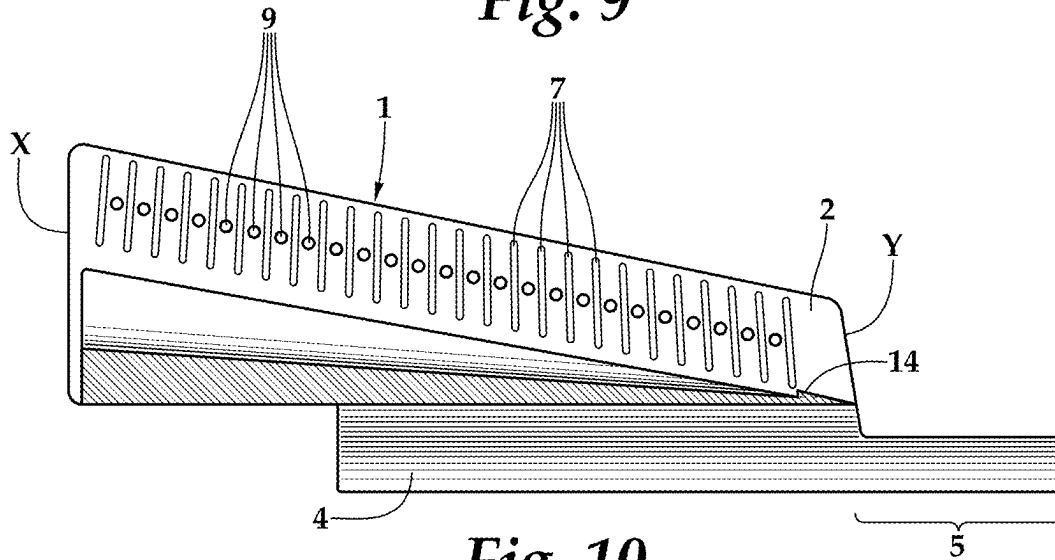


Fig. 10

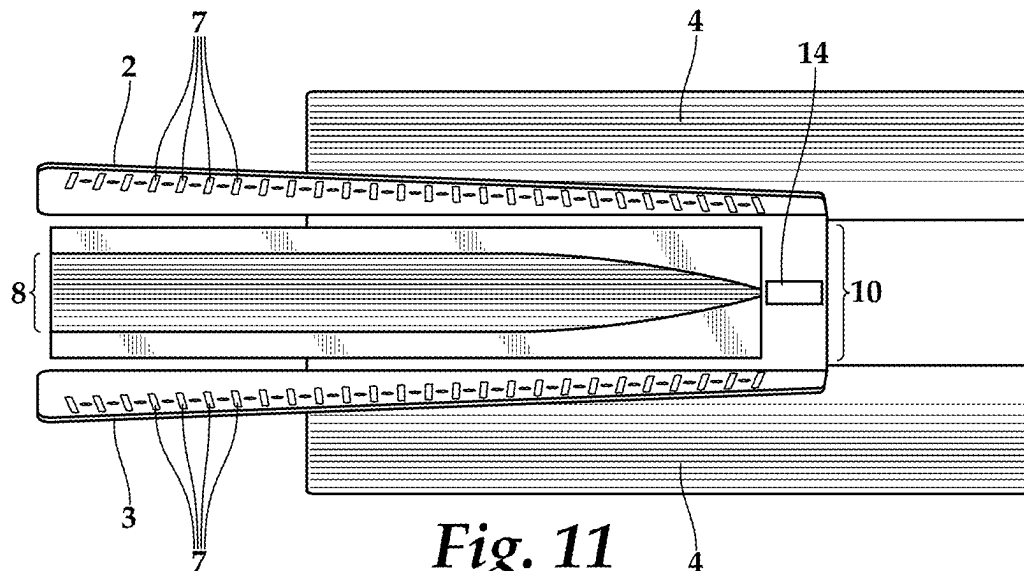
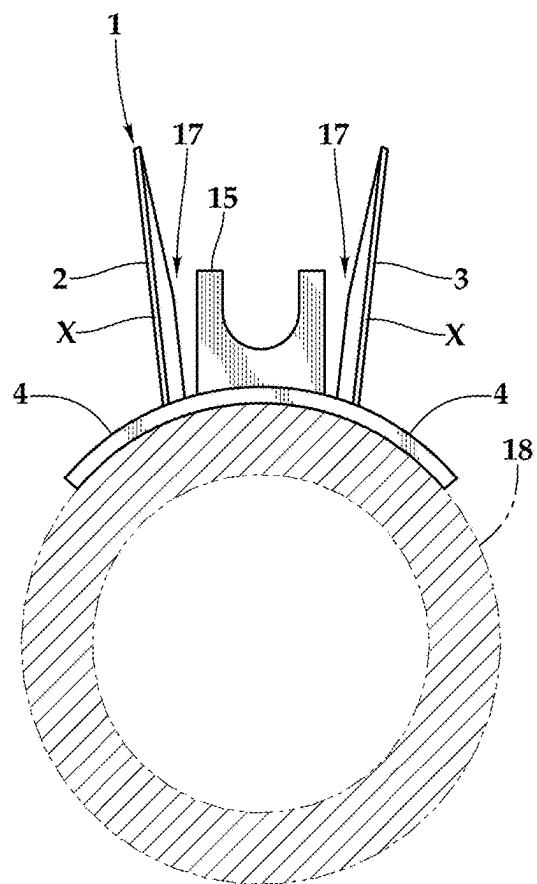
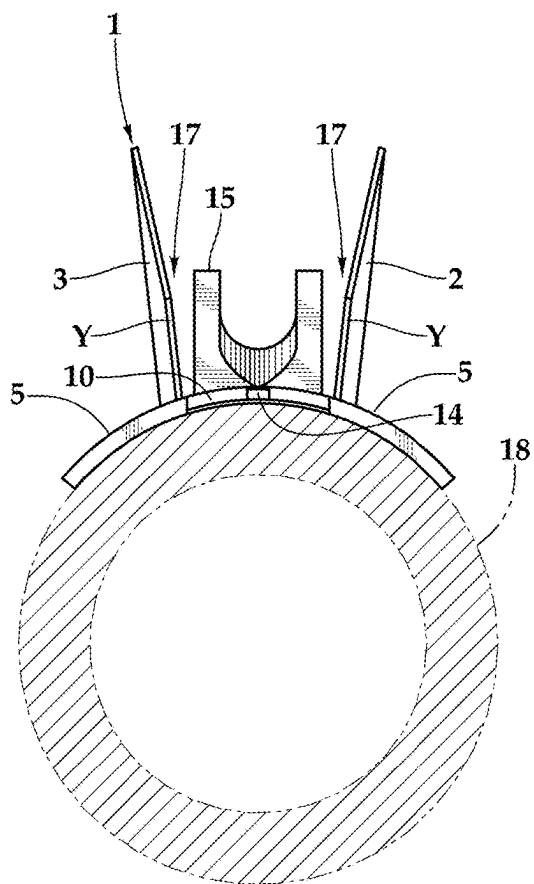
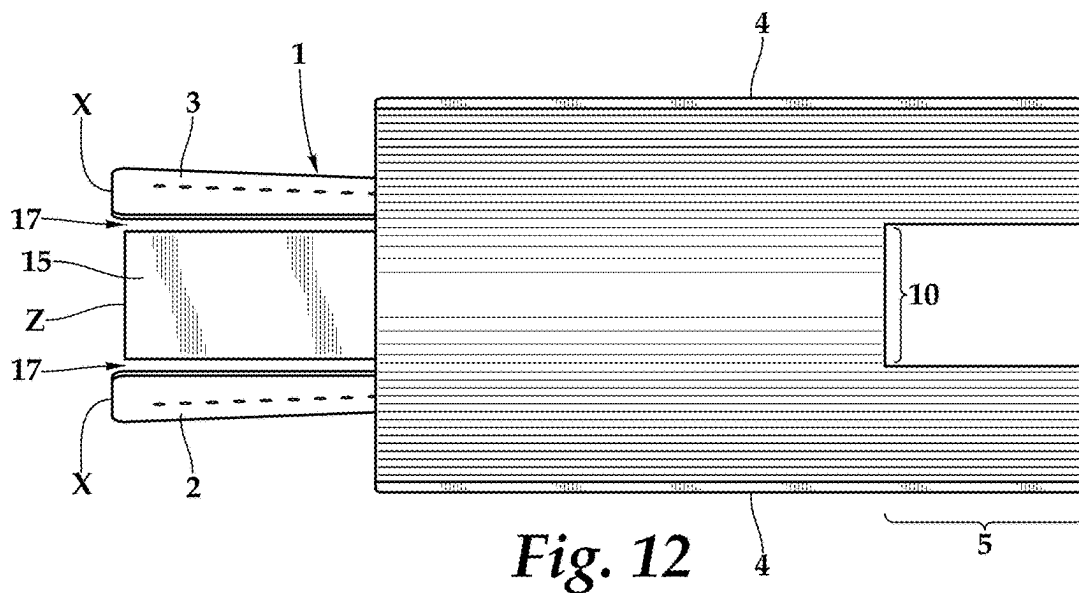


Fig. 11



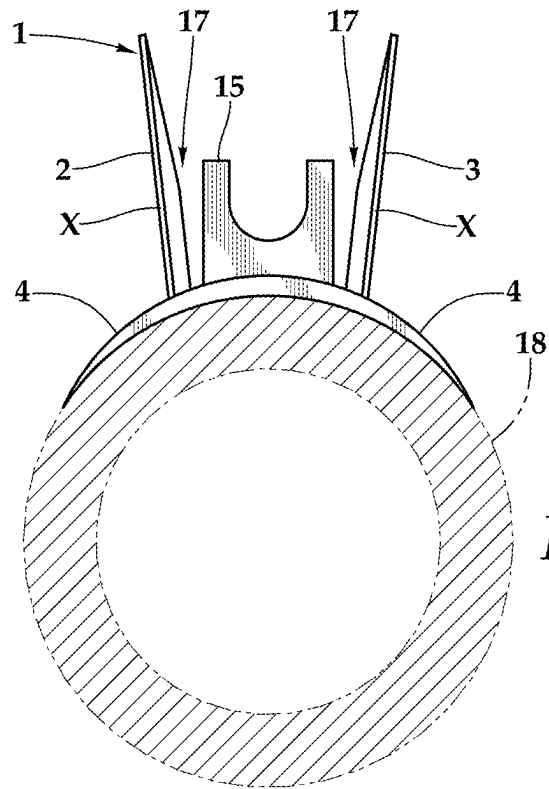


Fig. 15

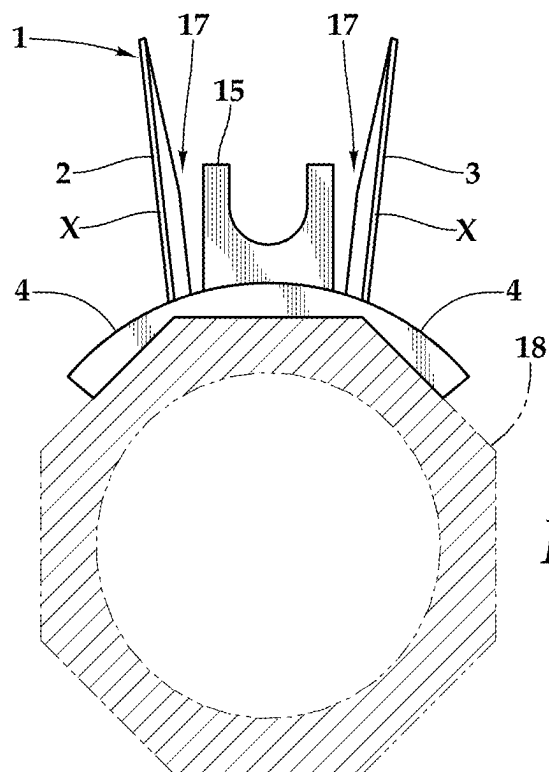


Fig. 16

REAR FIREARM SIGHT SLIDE ELEVATOR**CROSS-REFERENCE TO RELATED APPLICATION**

Pursuant to 35 U.S.C. § 119(e), this application claims priority back to U.S. Patent Application No. 63/634,778 filed on Apr. 16, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION**1. Field of the Invention**

The present invention relates generally to the field of firearm accessories, and more particularly, to a rear firearm sight slide elevator.

2. Description of the Related Art

Rear firearm sight elevators are used to adjust sight elevation settings by raising and lowering the rear sight of a firearm. The rear firearm sight step elevator is a commonly used type of rear firearm sight elevator. Conventional rear firearm sight step elevators are in the form of a single- or double-step rear sight elevator that is configured to raise and lower the rear sight of a firearm, thereby changing the sight elevation setting, using sight elevation steps. A single-step rear sight elevator has one stepping surface or “fin” (in the form of a staircase) that is used to raise the rear firearm sight. A double-step rear sight elevator has two parallel fins (increasing in height from rear to front) that are joined together by a base plate.

In single-step rear sight elevators, the elevator runs parallel to the barrel and down the center of the sight path of the rear sight of a firearm. This prevents single-step rear sight elevators from being used for higher sight elevation settings because the elevator would obstruct the sight path if it were made to be taller. Double-step rear sight elevators become exposed from under the rear sight of a firearm at certain settings, which makes them prone to unintentional lateral movements. This unintentional lateral movement can cause undesired sight elevation changes. Thus, there is a need for a rear firearm sight elevator that can provide higher sight elevation settings without obstructing the sight path or being prone to unintentional lateral movements.

Single- and double-step rear sight elevators have sight elevation steps that are difficult to force under the rear sight of a firearm. The rear sight of a firearm usually needs to be raised to relieve pressure from the rear firearm sight step elevator before the elevator can be moved to make sight elevation setting changes. This requires the operator to use both of his hands, thereby obstructing the operator’s view of the rear firearm sight step elevator and the rear sight of the firearm, and it often requires the operator to move the firearm from the shooting position, additionally complicating and slowing down the process of changing sight elevation settings.

Single- and double-rear sight step elevators have a small amount of exposed surface area to contact and grip, which makes them difficult to adjust. This small amount of exposed surface area only allows for contact by an operator’s fingertips. This makes maintaining an effective grip while adjusting a rear firearm sight step elevator extremely difficult, especially for operators with cold fingers, a weak grip, or when the operator is wearing gloves. Accordingly, sight elevation setting changes are slow and, for some operators,

nearly impossible. What is needed is a rear firearm sight elevator that (a) is easy to contact and grip; (b) is quickly, smoothly and easily adjusted without requiring the operator to lose sight of the rear firearm sight elevator and the rear sight of the firearm or to move the firearm out of a shooting position; and (c) without requiring a significant amount of physical pressure from the operator.

Single- and double-step rear sight elevators have little exposed surface area, which makes putting detailed and legible sight elevation reference markings on them unfeasible. As a result, an operator who is unfamiliar with the rear firearm sight elevator will be unable to change sight elevation settings quickly and effectively. For this reason, there is also a need for a rear firearm sight elevator that has a relatively large exposed surface area on which detailed and legible sight elevation reference markings may be displayed.

U.S. Pat. No. 3,975,851 (Benford, 1976) discloses a gun sight that is detachably mounted on the rear of the barrel. In one embodiment, the invention includes an elevation bar that is mounted directly underneath the sight to provide adjustable elevation of the sight. The elevation bar has steps and does not provide smooth elevation as in the present invention, which has a smooth ramp. The Benford invention also does not include fins or sight markings and is a whole sight system requiring gunsmithing to install. Unlike the present invention, it is not an improved elevator that can be installed by a lay person in two seconds.

U.S. Pat. No. 920,767 (Lindberg, 1909) provides a gun sight with a slide that is configured to raise or lower the height of the sight. This particular invention is an integrated sight-elevator system; it is not a separate slide elevator that can be used with an existing rear sight. It is structurally distinguishable from the present invention in that it does not incorporate fins or sight markings and is a whole sight system requiring gunsmithing to install, among other differences. The rear sight in the Lindberg invention is pivotally attached to the sight “body” or what is referred to in the present application as the sight “arm.” The rear sight itself is not part of the present invention, and the sight with which the present invention is used is not pivotally attached to the arm. The elevator of the Lindberg invention is similar to a single-step sight elevator without the steps.

U.S. Pat. No. 682,739 (Parsons et al., 1901) discloses an adjustable rear sight system that is semi-permanently affixed to the rifle. By contrast, the present invention is a rear sight slide elevator (not a rear sight system) that is easily detachable. The Parsons invention incorporates a pin or pivot that is configured to move the sight-leaf laterally relative to the rear sight via the longitudinal movement of the pin. It also includes a slide that is placed onto the leaf-spring. The present invention is one solid piece rather than several moving parts that require gunsmithing to be installed on a firearm, as in Parsons.

U.S. Pat. No. 670,012 (Brougher, 1901) provides a rear firearm sight that incorporates a double-step sight elevator. The elevation of the sight is adjusted via a dial-and-gear system that is not part of the present invention. None of the inventions discussed above offers the structural and functional advantages of the present invention, which is discussed more fully below.

BRIEF SUMMARY OF THE INVENTION

The present invention is a rear firearm sight slide elevator comprising: a base member; wherein the base member is configured to surround a part of a firearm barrel; and wherein the base member comprises a first wing on a first

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side of a front part of the base member and a second wing on a second side of a front part of the base member; a first fin and a second fin; wherein the first fin and the second fin are parallel to each other and situated at least in part on top of the base member; wherein the first fin comprises an inner wall and an outer wall, and the second fin comprises an inner wall and an outer wall; wherein either the inner wall of the first fin or the inner wall of the second fin comprises a plurality of vertically oriented ridges; and wherein the outer wall of at least one of the first fin or the second fin comprises a plurality of sight markings; and a ramp that is situated on a central portion of the base member between the first fin and the second fin; wherein the ramp has a smooth top edge and increases in height from a front end of the ramp to a rear end of the ramp; and wherein the ramp comprises a channel that is configured to provide a sight path from a user to a front sight on the firearm when the rear firearm sight slide elevator is installed on a firearm.

In a preferred embodiment, the invention further comprises a first gap between a first side of the ramp and the first fin and a second gap between a second side of the ramp and the second fin. In one embodiment, the base member is preferably concave in shape. In another embodiment, the base member is preferably configured to fit an octagonal firearm barrel. In a preferred embodiment, the base member comprises a first longitudinal edge along a first side of the base member and a second longitudinal edge along a second side of the base member; and the first longitudinal edge and the second longitudinal edge are both tapered.

In a preferred embodiment, the invention further comprises an aperture in between each of two adjacent ridges. Preferably, each of the first fin and the second fin comprises a front end and a rear end, and each of the first fin and the second fin increases in height from the front end of the fin to the rear end of the fin.

In a preferred embodiment, the invention further comprises a protrusion on top of a front end of the central portion of the base member in front of a front end of the ramp. Preferably, the channel tapers to a point directly behind the protrusion. A bottom surface of the ramp is preferably concave in shape to conform to an outside surface of a firearm barrel.

In a preferred embodiment, the invention is a rear firearm sight slide elevator comprising: a base member; wherein the base member is configured to surround a part of a firearm barrel; a first fin and a second fin; wherein the first fin and the second fin are parallel to each other and situated at least in part on top of the base member; wherein the first fin comprises an inner wall and an outer wall, and the second fin comprises an inner wall and an outer wall; wherein either the inner wall of the first fin or the inner wall of the second fin comprises a plurality of vertically oriented ridges; and wherein the outer wall of at least one of the first fin or the second fin comprises a plurality of sight markings; and a ramp that is situated on a central portion of the base member between the first fin and the second fin; wherein the ramp has a smooth top edge and increases in height from a front end of the ramp to a rear end of the ramp; and wherein the ramp comprises a channel that is configured to provide a sight path from a user to a front sight on the firearm when the rear firearm sight slide elevator is installed on a firearm.

In another preferred embodiment, the invention is a rear firearm sight slide elevator comprising: a base member; wherein the base member is configured to surround a part of a firearm barrel; and wherein the base member comprises a first wing on a first side of a front part of the base member and a second wing on a second side of a front part of the base

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member; a first fin and a second fin; wherein the first fin and the second fin are parallel to each other and situated at least in part on top of the base member; wherein the first fin comprises an inner wall and an outer wall, and the second fin comprises an inner wall and an outer wall; and wherein either the inner wall of the first fin or the inner wall of the second fin comprises a plurality of vertically oriented ridges; and a ramp that is situated on a central portion of the base member between the first fin and the second fin; wherein the ramp has a smooth top edge and increases in height from a front end of the ramp to a rear end of the ramp; and wherein the ramp comprises a channel that is configured to provide a sight path from a user to a front sight on the firearm when the rear firearm sight slide elevator is installed on a firearm.

In yet another preferred embodiment, the invention is a rear firearm sight slide elevator comprising: a base member; wherein the base member is configured to surround a part of a firearm barrel; and a first fin and a second fin; wherein the first fin and the second fin are parallel to each other and situated at least in part on top of the base member; wherein the first fin comprises an inner wall and an outer wall, and the second fin comprises an inner wall and an outer wall; and wherein either the inner wall of the first fin or the inner wall of the second fin comprises a plurality of vertically oriented ridges; and a ramp that is situated on a central portion of the base member between the first fin and the second fin; wherein the ramp has a smooth top edge and increases in height from a front end of the ramp to a rear end of the ramp; and wherein the ramp comprises a channel that is configured to provide a sight path from a user to a front sight on the firearm when the rear firearm sight slide elevator is installed on a firearm.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a rear perspective view of the present invention shown with the rear firearm sight slide elevator installed on a firearm barrel.

FIG. 2 is a side view of a rifle with a prior art single-step rear firearm sight elevator installed on the firearm.

FIG. 3 is a side view of a rifle with the rear firearm sight slide elevator of the present invention installed on the firearm.

FIG. 4 is a detail side view of a prior art single-step rear firearm sight elevator installed on the firearm.

FIG. 5 is a detail side view of the rear firearm sight slide elevator of the present invention shown in relation to the rear sight of the firearm prior to installation.

FIG. 6 is a detail side view of the rear firearm sight slide elevator of the present invention shown in relation to the rear sight on the firearm after installation.

FIG. 7 is a detail section view of the present invention installed on the firearm with the rear sight in a first position in relation to the rear firearm sight elevator.

FIG. 8 is a detail section view of the present invention installed on the firearm with the rear sight in a second position in relation to the rear firearm sight elevator.

FIG. 9 is a side view of the present invention shown.

FIG. 10 is a longitudinal section view of the present invention.

FIG. 11 is a top view of the present invention.

FIG. 12 is a bottom view of the present invention.

FIG. 13 is a front view of the present invention shown in relation to a circular firearm barrel.

FIG. 14 is a rear view of the present invention shown in relation to a circular firearm barrel.

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FIG. 15 is a rear view of an alternate embodiment of the present invention in which the outer edges of the base member are tapered.

FIG. 16 is a rear view of an alternate embodiment of the present invention in which the base member is shaped to conform to an octagonal firearm barrel.

REFERENCE NUMBERS

- 1 Rear firearm sight slide elevator
- 2 First fin
- 3 Second fin
- 4 Base member
- 5 Wings (of base member)
- 6 Rear firearm sight (not part of the present invention)
- 7 Ridges
- 9 Channel (in ramp)
- 9 Apertures
- 10 Central portion (of base member)
- 11 Arm
- 12 Bracket
- 13 Single-step rear sight elevator (prior art)
- 14 Bump (on front end of central portion of base member)
- 15 Ramp (on central portion of base member)
- 16 Sight markings
- 17 Gap
- 18 Firearm barrel

DETAILED DESCRIPTION OF INVENTION

FIG. 1 is a rear perspective view of the present invention shown with the rear firearm sight slide elevator installed on a firearm barrel. As shown in this figure, the present invention 1 comprises a first fin 2 and a second fin 3. These fins 2, 3 are parallel to each other and are situated on top of a concave-shaped base member 4. The base member 4 is configured to surround part of the firearm barrel. Part of the bottom edge of the first fin 2 is connected to one side of the top of the base member 4, and part of the bottom edge of the second fin 3 is connected to the other side of the top of the base member 4. In a preferred embodiment, at least half of the bottom edge of each fin 2, 3 is connected to the top of the base member 4. This base member 4 is comprised of two forward wings 5 and a central portion 10. The central portion of the base member 4 underlies the bottom edges of the first and second fins 2, 3 for at least half the length of the fin.

For reference, the proximal end of each fin (that is, that part of the fin that is closest to the stock of the firearm) is marked as "X" (see also FIG. 12), and the distal end of each fin (that is, that part of the fin that is closest to the muzzle of the firearm) is marked as "Y" in FIG. 1. In a preferred embodiment, the two fins 2, 3 are of equal length, and the base member 4 is approximately as long as the fins. Note that the wings 5 of the base member 4 extend past the distal end Y of the fin. As noted above, the base member 4 is configured to hug the top part of the barrel 18, and the fins 2, 3 are situated atop the base member 4. The purpose of the base member 4 is not only to support the ramp 15 but also (together with the wings 5) to reduce any lateral movement of the sight slide elevator 1 on the firearm barrel.

In a preferred embodiment, the degree of concavity of the base member 4 is preferably approximately the same as the degree of convexity of the outside of the firearm barrel. A ramp 15 is situated on top of the central portion 10 of the base member 4. This ramp 15 is smooth (i.e., it does not have steps) and increases in height from the distal end of the ramp (closest to the firearm muzzle) to the proximal end of the

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ramp (closest to the firearm stock). Similarly, the height of each fin increases from the distal end to the proximal end of each fin; however, the height of the ramp is always less than the height of each fin. Also, the height of each fin increases at the same rate as the height of the other fin so that both fins are equal in height at all points along the length of the fins. The ramp 15 comprises a channel 8, the purpose of which is to provide a sight path through the ramp 15.

In a preferred embodiment, there is a bump or protrusion 14 on top of the base member 4 at the distal (or front) end of the central portion 10 of the base member directly in front of the distal (or front) end of the ramp 15. The purpose of this bump 14 is to secure the rear firearm sight 6 in a first position; as manual pressure is applied to the sight elevator in a distal direction (pushing the sight elevator underneath the sight and toward the muzzle end of the barrel), the sight moves up and onto the ramp. As further pressure is applied, the sight moves further up the ramp, thereby increasing the height of the sight. The proximal ends of the fins 2, 3 provide a relatively large surface for the user's fingers and/or thumb to manipulate the sight elevator.

In a preferred embodiment, each fin 2, 3 comprises a plurality of parallel ridges 7 spaced equally apart from one another on the inside surface of the fin. These parallel ridges 7 are preferably situated on a top part of the inside surface of the fin above the base member 4 and above the ramp 15. In between each set of adjacent parallel ridges is a small, preferably circular, aperture 9. The apertures 9 on both fins are aligned laterally so that when the rear sight 6 is situated between two pairs of adjacent ridges 7 on either side of the rear sight 6, the aperture 9 on either fin is blocked by the rear sight 6. In this manner, the operator can tell where the rear sight is simply by viewing the side of either fin. Optionally, the first two ridges 7 (on the distal end of the sight elevator, proximate to the bump 14) are thicker than the rest of the ridges 7. As noted above, the base member 4 (except for the wings 5) is situated underneath the two fins 2, 3 and has a concave bottom surface (marked as "Z" in FIG. 1) that matches the curvature of the outside of the barrel. The bottom surface of the ramp may also be concave to conform to the shape of the firearm barrel.

FIG. 2 is a side view of a rifle with a prior art single-step rear firearm sight elevator installed on the firearm, and FIG. 3 is a side view of a rifle with the rear firearm sight slide elevator of the present invention installed on the firearm. The purpose of these two figures is to illustrate that the present invention provides a greater surface area for gripping the fins of the sight elevator than the prior art version shown in FIG. 2. These two figures also show where the sight elevator is located on the firearm. The present invention is typically used with all rifles that have a leaf-spring-mounted rear sight as depicted in FIG. 1 (examples of such rifles includes, but are not limited to, the Savage Model 62 and the Sport King Carbine Model A-102).

FIG. 4 is a detail side view of a prior art single-step rear firearm sight elevator installed on the firearm. As shown in this figure, the rear sight 6 is connected to an arm 11 that is secured to the barrel by a bracket 12. The arm 11 is configured to hold the rear sight 6 flat against the barrel, but the single-step sight elevator 13 can be used to raise the sight by varying degrees. The single-step sight elevator 13 shown in this figure is inserted into a slot (not shown) in the arm 11 and then slid rearward until the sight 6 is at the desired level on the sight elevator 13. The further rearward the sight elevator is slid, the further the sight is raised. The arm 11 acts as a leaf spring to hold the sight 6 in place on the sight

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elevator, which includes a number of steps (as described above). With each successive step, the sight is raised higher.

FIG. 5 is a detail side view of the rear firearm sight slide elevator of the present invention shown in relation to the rear sight of the firearm prior to installation, and FIG. 6 is a detail side view of the rear firearm sight slide elevator of the present invention shown in relation to the rear sight on the firearm after installation. As shown in these two figures, the wings 5 are first positioned on either side of the front end of the arm 11 with the sight 6 abutting up against the front end of the base member 4. Manual pressure is then applied to the sight elevator 1, and the elevator is pushed in a distal direction (toward the muzzle of the firearm). This pressure causes the rear sight 6 to pass over the bump 14 and up onto the ramp 15. As further pressure is applied by the operator in a distal direction, the sight rises further up on the ramp 15, thereby raising the position of the sight 6 relative to the front firearm sight, causing the operator to raise the muzzle end of the firearm to maintain alignment of the front and rear sights. It should be noted that during this entire procedure, the operator's view of the front and rear firearm sights is never blocked by the sight elevator. Note that the apertures 9 are preferably configured in a line that is parallel to the top surface of the fin and, therefore, at a diagonal angle relative to the longitudinal axis of the firearm.

FIG. 7 is a detail section view of the present invention installed on the firearm with the rear sight in a first position in relation to the rear firearm sight elevator. In this figure, the muzzle end of the firearm (not shown) is to the right, and the firearm stock (not shown) is to the left. This figure shows the sight 6 in relation to the first fin 2 after the sight 6 has cleared the bump 14 on the distal end of the base member 4 at the bottom of the ramp 15. As shown, the lateral edge of the sight 6 is situated between two of the ridges 7 and is blocking one of the apertures 9.

FIG. 8 is a detail section view of the present invention installed on the firearm with the rear sight in a second position in relation to the rear firearm sight elevator. In this figure, the elevator 1 has been pushed manually by the operator in a distal direction (toward the muzzle or to the right in FIG. 8). As this happens, the sight 6 rides higher up on the ramp 15 and is raised relative to the front sight (not shown), which requires the operator to raise the muzzle end of the firearm in order to maintain alignment of the front and rear sights.

FIG. 9 is a side view of the present invention. As shown in this figure, the invention preferably includes sight markings 16 on the outside surface of at least one of the fins 2, 3 directly underneath at least some of the plurality of apertures 9. Thus, each fin must have sufficient surface area to display the sight markings. The purpose of the sight markings is to assist the operator in determining how far up the ramp 15 to push the sight 6 to achieve the desired shooting range.

FIG. 10 is a longitudinal section view of the present invention. This is the same view as is shown in FIG. 7 except that the rear sight and firearm have been omitted.

FIG. 11 is a top view of the present invention. This figure clearly shows the bump 14 on the distal end of the central portion 10 of the base member 4. As shown in this figure, the channel 8 preferably tapers to a point directly behind (on the proximal side of) the bump 14. This figure also shows that the two fins 2, 3 are preferably splayed slightly in an outward direction from bottom to top. This enables the user to obtain a bit of flex of the fins when gripping them. Note also that the fins 2, 3 terminate at or near the front edge of the central portion 10 of the base member 4.

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FIG. 12 is a bottom view of the present invention. As shown here, the wings 5 are of equal length, and both the fins 2, 3 and the ramp 15 extend past the proximal (or rear) end of the base member 4 (see also FIG. 3) for a distance that is equal to approximately one-third ($\frac{1}{3}$) of the entire length of the fins and approximately one-half ($\frac{1}{2}$) of the length of the central portion of the base member. (In this figure, "proximal" is to the left, and "distal" is to the right.) Note that there is preferably a gap 17 on either side of the ramp 15 between the outside walls of the ramp and the fins 2, 3 on either side of the ramp. The presence of this gap 17 allows the operator to squeeze the fins together when gripping the sight elevator or to fit the user's thumb knuckle in between the two fins in order to separate them apart to slide the sight elevator backwards (toward the user).

FIG. 13 is a front view of the present invention shown in relation to a firearm barrel, and FIG. 14 is a rear view of the present invention shown in relation to a firearm barrel. These two figures illustrate the near-perfect match between the curvature of the firearm barrel 18 and the curvature of the base member 4. They also show the slight outward splaying of the fins 2, 3 and the gap 17 between the ramp 15 and the fins. As shown in these two figures, the overall cross-sectional thickness of the ramp 15 preferably increases from front to back. (As used here, the term "front" means closest to the muzzle, and the term "back" means closest to the stock.) This is so that the back part of the ramp 15 (which is not supported by the base member 4) has sufficient structural integrity while providing elevation to the rear firearm sight. When the invention is in use, the operator would grip the rear part of the elevator 1 at the fins 2, 3, position the elevator on the firearm as shown in FIG. 5, and push the elevator forward (toward the muzzle end of the firearm) until the desired position is reached. This can be done while holding the firearm in a shooting position or with the firearm in a non-shooting position with reference to the sight markings.

FIG. 15 is a rear view of an alternate embodiment of the present invention in which the outer edges of the base member are tapered. In this embodiment, the outer edges of the base member 4 are tapered, as shown, to provide greater flex against the firearm barrel.

FIG. 16 is a rear view of an alternate embodiment of the present invention in which the base member is shaped to conform to an octagonal firearm barrel. In this embodiment, the top of the base member 4 is still convex in shape (as shown in FIG. 14), but the bottom of the base member is octagonal in shape (while still maintaining a general concavity) to conform to an octagonal firearm barrel. In a preferred embodiment, the bottom of the base member is configured (shaped) to fit the barrel shape and dimensions.

The present invention has several advantages over the prior art, including the fact that it does not require the use of both hands to adjust sight elevation settings. Prior art rear sight elevators require an operator to lift the rear sight with one hand and then move the rear firearm sight step elevator with the other. The present invention can be slid with one hand and is easy to handle and grip.

The present invention also allows an operator to change the elevation setting of a rear firearm sight more quickly with a single action of one hand, sliding the elevator forward or backward. Prior art rear sight elevators required the operator to stabilize the firearm and then lift the rear sight with one hand while moving the rear firearm sight step elevator with the other hand. The present invention can be installed in less than two seconds without tools by holding

the firearm with one hand and sliding the elevator under the rear sight with the other hand.

Other advantages of the present invention include the fact that it provide easily readable detailed sight elevation reference markings on the side of the elevator. In addition, it allows the operator to achieve a relatively high sight setting without obstruction the operator's view of the front sight. Prior art sight elevators only allow for short to medium settings before the front sight becomes obscured or the elevator becomes unstable.

The present invention enables an operator to adjust the sight elevation even with a weak grip, cold fingers, or while wearing gloves. Prior art step sight elevators can only be manipulated with a significant amount of focused fingertip pressure, and the parts are very small.

Lastly, the concave shape of the base member and the wings enables the present invention to hug or grip the firearm barrel while still being able to slide, providing greater stability to the sight elevator, which in turn translates to greater accuracy in terms of the overall performance of the firearm.

Although the preferred embodiment of the present invention has been shown and described, it will be apparent to those skilled in the art that many changes and modifications may be made without departing from the invention in its broader aspects. The appended claims are therefore intended to cover all such changes and modifications as fall within the true spirit and scope of the invention.

I claim:

1. A rear firearm sight slide elevator comprising:

(a) a base member;

wherein the base member is configured to surround a part of a firearm barrel; and

wherein the base member comprises a first wing on a first side of a front part of the base member and a second wing on a second side of a front part of the base member;

(b) a first fin and a second fin;

wherein the first fin and the second fin are parallel to each other and situated at least in part on top of the base member;

wherein the first fin comprises an inner wall and an outer wall, and the second fin comprises an inner wall and an outer wall;

wherein either the inner wall of the first fin or the inner wall of the second fin comprises a plurality of vertically oriented ridges; and

wherein the outer wall of at least one of the first fin or the second fin comprises a plurality of sight markings; and

(c) a ramp that is situated on a central portion of the base member between the first fin and the second fin;

wherein the ramp has a smooth top edge and increases in height from a front end of the ramp to a rear end of the ramp; and

wherein the ramp comprises a channel that is configured to provide a sight path from a user to a front sight on the firearm when the rear firearm sight slide elevator is installed on a firearm.

2. The rear firearm sight slide elevator of claim 1, further comprising a first gap between a first side of the ramp and the first fin and a second gap between a second side of the ramp and the second fin.

3. The rear firearm sight slide elevator of claim 1, wherein the base member is concave in shape.

4. The rear firearm sight slide elevator of claim 1, wherein the base member is configured to fit an octagonal firearm barrel.

5. The rear firearm sight slide elevator of claim 1, wherein the base member comprises a first longitudinal edge along a first side of the base member and a second longitudinal edge along a second side of the base member; and

wherein the first longitudinal edge and the second longitudinal edge are both tapered.

6. The rear firearm sight slide elevator of claim 1, further comprising an aperture in between each of two adjacent ridges.

7. The rear firearm sight slide elevator of claim 1, wherein each of the first fin and the second fin comprises a front end and a rear end, and each of the first fin and the second fin increases in height from the front end of the fin to the rear end of the fin.

8. The rear firearm sight slide elevator of claim 1, further comprising a protrusion on top of a front end of the central portion of the base member in front of a front end of the ramp.

9. The rear firearm sight slide elevator of claim 8, wherein the channel tapers to a point directly behind the protrusion.

10. The rear firearm sight slide elevator of claim 1, wherein a bottom surface of the ramp is concave in shape to confirm to an outside surface of a firearm barrel.

11. A rear firearm sight slide elevator comprising:

(a) a base member:

wherein the base member is configured to surround a part of a firearm barrel;

(b) a first fin and a second fin;

wherein the first fin and the second fin are parallel to each other and situated at least in part on top of the base member;

wherein the first fin comprises an inner wall and an outer wall, and the second fin comprises an inner wall and an outer wall;

wherein either the inner wall of the first fin or the inner wall of the second fin comprises a plurality of vertically oriented ridges; and

wherein the outer wall of at least one of the first fin or the second fin comprises a plurality of sight markings; and

(c) a ramp that is situated on a central portion of the base member between the first fin and the second fin;

wherein the ramp has a smooth top edge and increases in height from a front end of the ramp to a rear end of the ramp; and

wherein the ramp comprises a channel that is configured to provide a sight path from a user to a front sight on the firearm when the rear firearm sight slide elevator is installed on a firearm.

12. A rear firearm sight slide elevator comprising:

(a) a base member;

wherein the base member is configured to surround a part of a firearm barrel; and

wherein the base member comprises a first wing on a first side of a front part of the base member and a second wing on a second side of a front part of the base member;

(b) a first fin and a second fin;

wherein the first fin and the second fin are parallel to each other and situated at least in part on top of the base member;

wherein the first fin comprises an inner wall and an outer wall, and the second fin comprises an inner wall and an outer wall; and

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wherein either the inner wall of the first fin or the inner wall of the second fin comprises a plurality of vertically oriented ridges; and

- (c) a ramp that is situated on a central portion of the base member between the first fin and the second fin; 5

wherein the ramp has a smooth top edge and increases in height from a front end of the ramp to a rear end of the ramp; and

wherein the ramp comprises a channel that is configured to provide a sight path from a user to a front sight on the firearm when the rear firearm sight slide elevator is installed on a firearm. 10

13. A rear firearm sight slide elevator comprising:

- (a) a base member; 15

wherein the base member is configured to surround a part of a firearm barrel; and

- (b) a first fin and a second fin;

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wherein the first fin and the second fin are parallel to each other and situated at least in part on top of the base member;

wherein the first fin comprises an inner wall and an outer wall, and the second fin comprises an inner wall and an outer wall; and

wherein either the inner wall of the first fin or the inner wall of the second fin comprises a plurality of vertically oriented ridges; and

- (c) a ramp that is situated on a central portion of the base member between the first fin and the second fin;

wherein the ramp has a smooth top edge and increases in height from a front end of the ramp to a rear end of the ramp; and

wherein the ramp comprises a channel that is configured to provide a sight path from a user to a front sight on the firearm when the rear firearm sight slide elevator is installed on a firearm.

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